

## SUBMISSION

# Plinth — React App Development with AI

Week 3 / Frontend AI Engineering / Track Assignment

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## The Application

Plinth is an open-source platform where developers create gallery-style exhibit pages for their projects. Built with Next.js 16, React 19, Tailwind CSS v4, and TypeScript. 8 routes, responsive across 375px / 768px / 1280px, accessibility-first.

**Live preview** [plinth-cyan.vercel.app](https://plinth-cyan.vercel.app)

**Repository** [github.com/ZAYNINFINITY/flyrank-ai-internship](https://github.com/ZAYNINFINITY/flyrank-ai-internship) (root: `week-03/app/`)

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## Prompts Used During Development

12 sequential prompts. Phase A (1–3) was documentation only. Phase B (4–12) was the full code build.

### PHASE A — DOCUMENTATION

• Prompt 1 **Through-Line doc**

Create the one-line claim, the full content map table (route, purpose, CTA), and an honest "still need to gather" list.

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• Prompt 2 **Identity Kit doc**

Choose fonts (Space Grotesk + Inter), define the 3-color palette with hex and OKLCH values, describe the favicon concept, write the two-line style note.

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• Prompt 3 **Curate Images**

List every image the site needs mapped to specific routes, mark each as real or generated, write rejection notes explaining why certain images were declined.

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### PHASE B — CODE BUILD

• Prompt 4 **Scaffold**

Initialize Next.js 16 with Tailwind v4. Set up `@theme inline` with OKLCH color tokens. Create empty route shells for all 8 pages.

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• Prompt 5 **Base Components**

Build 5 atomic primitives: Frame, Spotlight Button, Ghost Button, Museum Tag Label, Floor Directory. All with 44×44px tap targets and focus-visible states.

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• Prompt 6 **Landing Page**

3 full-viewport beats: entrance, exhibit concept with preview frames, explore CTA. Nav with "Menu" collapse on mobile.

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• Prompt 7 **Explore Page**

Explore grid with live slots (LIVE tag) and opening soon slots (gray tag, aria-disabled). Empty-state copy. Mock data.

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• Prompt 8 **Exhibit Page**

Room 0 entrance (name, role, avatar, Floor Directory) and Rooms 1–4 alternating image-left/right with real project copy and stack labels.

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• Prompt 9 **Remaining Routes**

Dashboard, Login, Health, About, and 404 pages — all with honest placeholder content.

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• Prompt 10 **Responsive Pass**

Audit at 375px, 768px, 1280px. Standardize gutters (24→32→48px). Mobile nav overlay. Horizontal scrollable Floor Directory.

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• Prompt 11 **Accessibility + Motion**

`prefers-reduced-motion`, skip-to-content link, Escape key dismissal, body scroll lock, `role="dialog"`, `aria-modal`, `autoComplete` on inputs.

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• Prompt 12 **Repo Hygiene**

README, MIT LICENSE, CONTRIBUTING.md, `.env.example`, updated package.json metadata.

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## How AI Assisted

- **Architecture** — file structure, component hierarchy, data flow. Separated raw shadcn primitives from Plinth-specific components.
- **Code generation** — all page layouts, component implementations, mock data, and route configurations from text prompts.
- **Design tokens** — converted hex palette ( `#0F172A`, `#FAFAFA`, `#2563EB` ) to Tailwind v4 OKLCH format.
- **Accessibility** — focus-visible outlines, `aria-disabled`, `role="dialog"`, `prefers-reduced-motion`.
- **Documentation** — generated all three Phase A docs from the technical brief.

The workflow was prompt-driven. I described intent, AI generated code, I reviewed output, we iterated. Each prompt built on the confirmed output of the previous one.

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## Manual Improvements

## 2 OKLCH Color Correction

AI generated incorrect OKLCH values for the accent blue. Manually verified against Tailwind's documented blue-600 ( `oklch(0.546 0.245 262.881)` ) and corrected the `@theme inline` block. Wrong values would have broken every accent element on the site.

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## 2 Contrast Ratio Verification

AI estimated contrast without calculating. Manually computed all pairs using OKLCH luminance: text on background 17.11:1, accent on background 4.95:1, white on accent 5.17:1. These determined which opacity levels are acceptable for different UI roles.

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## 3 Mobile Navigation Rebuild

Replaced AI's hamburger icon with a text-only "Menu" trigger. Added Escape key dismissal, auto-focus, and body scroll lock — none of which were in the original output.

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## 3 Floor Directory Mobile Behavior

Original component was vertical-only. Modified to horizontal scrollable strip on mobile, vertical on desktop — a responsive adjustment AI didn't anticipate.

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## 3 Gather List Accuracy

Updated the through-line doc's "Still Need to Gather" section to cross off items already completed during the build, ensuring docs reflect actual project state.

Built with AI assistance. Verified and corrected manually. Week 3 of a public build-in-progress.